Department of Software Engineering
Government College University, Faisalabad

Lecture Plan

Course Title	Application of Information & Communication Technologies		
Course Code	CGE-301	**Credit Hours**	3 (2 – 1)
Class	BS(SE)	**Semester**	1st

Week	Lecture	Topic	Source Book / Notes	Lab /Tutorials
1	1.1	Introduction to Computers and Computing History	Chapter 1: "Introduction"	Understand Physical parts of computer
	1.2	- Brief history of computers		
	1.3	- Generations of computers		
2	2.1	Computer Components and Architecture	Chapter 2: "Computer Organization"	Hardware Components of Motherboard
	2.2	- Processor, Memory, and Hardware		
	2.3	- Software and Types of Software		
3	3.1	Number Systems and Conversions	Chapter 6: "Data Representation"	Installation of Operating System
	3.2	- System Software: Importance and Types		
	3.3	Number Systems and Conversions		
4	4.1	Types of Computers	Chapter 8: "Personal Computers"	Downloading and Installation of Software Packages
	4.2	- Super, Mainframe, Mini, and Micro Computers		
	4.3	- Introduction to CBIS (Computer-Based Information System)		
		Using MS Word for documentation		
5	5.1	Computer Codes	Chapter 5: "Computer codes and Boolean Algebra"	Formatting of text, paragraphs, table of contents in MS Word
	5.2	Computer Arithmetic		
	5.3	Boolean Algebra and Logic Circuits		
6	6.1	Processor and Memory Types	Chapter 6: "Processor and Memory"	Table of contents, Styles and various controls in MS Word
	6.2	- Architecture of CPU and types		
	6.3	- Primary and Secondary Memory and types		
7	7.1	Input VS Output Devices	Chapter 7: "Input and Output Devices"	Inserting tables, shapes, forms, symbols in MS Word
	7.2	- Input devices, Pointing Devices, Vision and Draw devices		
	7.3	- Output Devices, Printers, Plotters		
8	8.1	Primary Vs Secondary Storage, Magnetic tapes, Magnetic Disks, Cache memory, Virtual memory	Chapter 8: "Primary Secondary Storage Devices"	Managing Page Layouts, Watermarks, page setting and margins in MS Word
	8.2	- Working of Hard Disk and its architecture		
	8.3	SSD, NVM, RAD, Optical Disk, Portable devices and Flash Disks		
		Mid-Exam		
9	9.1	Data Communications and Networks	Chapter 9: "Data communication and networks"	Mail Merge, referencing, spelling and grammar checking in MS Words
	9.2	Transmission modes, types of networks, mediums		
	9.3	Network Topologies		

10	10.1	Data communication mediums	Chapter 9: "Data communication and networks"	Securing MS Word Document and concluding MS Word
	10.2	Internet and web services		
	10.3	IP addressing and networking		
11	11.1	Algorithms and Flowcharts	Chapter 11: "Planning the Computer Program"	MS Power Point, Installation, Formatting and Slide designing
	11.2	Planning of Computer programs		
	11.3	Flowcharts and Pseudocodes		
12	12.1	Operating Systems	Chapter 12: "Operating Systems"	Slide Transition and Animation text in MS Power Point
	12.2	Functions of Operating Systems		
	12.3	CPU Scheduling, Process management and various types of Operating systems		
13	13.1	Internet and Web	Chapter 13: "The Internet"	Finalizing MS Power Point
	13.2	- Challenges and Outcomes of Internet Era		
	13.3	Internet and web services, Protocols, Tools, Modern Software		
14	14.1	Classification of Computers	Chapter 14: "Classification of Computers"	MS Excel, Formulas, Data Manipulations
	14.2	Notebook, Workstation, PC, Mainframes, Super Computer, Server Machines		
	14.3	Smart devices		
15	15.1	Modern Software Tools and Technologies		Networking Basics, Cabling, Connections
	15.2	Market trends and devices		
	15.3	Innovation in the world		
16	16.1	Project and Course Review		Intro to MS Visio, MS Publisher, Internet tools and tricks
	16.2	Discussions		
	16.3	Conclusions		
		- Final Exam		

d. Attendance Record